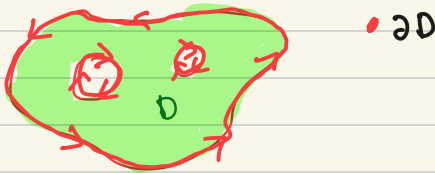


April 29, 2026

e.g. Region D in $\mathbb{R}^2$ with simple closed boundary



Theorem (Green's Theorem): Suppose D is a compact region in $\mathbb{R}^2$ with boundary $C = \partial D$ consisting of finitely many simple closed piecewise diff. curves.

Give C an orientation such that if we travel C following the orientation, the interior of the region is always to the left.

Suppose $\vec{F}(x,y) = (P(x,y), Q(x,y))$ is a C^1 vector field, then

$$\oint_C P(x,y)dx + Q(x,y)dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy.$$

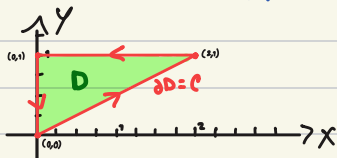
→ a.k.a. Divergence Theorem in $\mathbb{R}^2$
Vector formulation of Green's Thm:

Same hypothesis. Let $\hat{n}(s)$ be the unit outward normal vector to the region D at a point s on the boundary C . Then,

$$\oint_C \vec{F} \cdot \hat{n} ds = \iint_D \nabla \cdot \vec{F} dx dy.$$

$$\begin{aligned} \nabla \cdot \vec{F} dx dy &= \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx dy \\ \vec{F} \cdot \hat{n} ds &= -Q dx + P dy \end{aligned}$$

e.g. Let's see Green's Theorem in action for
 D in the triangular region with vertices $(0,0)$, $(2,1)$, $(0,1)$,
 $C = \partial D$ and $\vec{F}(x,y) = (2x+y, x^2)$.



1) Vector line integral:

parametrize C with the orientation
 leaving D to the left

$$\vec{\gamma}(t) = \begin{cases} (t, \frac{1}{2}t) & , 0 \leq t < 2 \\ (4-t, 1) & , 2 \leq t < 4 \\ (0, 5-t) & , 4 \leq t \leq 5 \end{cases}$$

$$\vec{v}(t) = \vec{\gamma}'(t) = \begin{cases} (1, \frac{1}{2}) & , 0 \leq t < 2 \\ (-1, 0) & , 2 \leq t < 4 \\ (0, -1) & , 4 \leq t \leq 5 \end{cases}$$

$$\vec{F}(\vec{\gamma}(t)) = \begin{cases} (2t + \frac{1}{2}, t^2) = (\frac{4t+1}{2}, t^2) & , 0 \leq t < 2 \\ (9-2t, (4-t)^2) & , 2 \leq t < 4 \\ (5-t, 0) & , 4 \leq t \leq 5 \end{cases}$$

$$\oint_C \vec{F} \cdot d\vec{s} = \int_0^2 \left(\frac{4t+1}{2}, t^2\right) \cdot \left(1, \frac{1}{2}\right) dt + \int_2^4 (9-2t, (4-t)^2) \cdot (-1, 0) dt + \int_4^5 (5-t, 0) \cdot (0, -1) dt$$

$$= \int_0^2 \left(\frac{4t+1}{2} + \frac{t^2}{2}\right) dt + \int_2^4 (9-2t) dt + \int_4^5 (5-t) dt$$

$$= \left[\frac{2t^2+t}{2} + \frac{t^3}{6}\right]_{t=0}^{t=2} + \left[9t - t^2\right]_{t=2}^{t=4} + \left[5t - \frac{t^2}{2}\right]_{t=4}^{t=5}$$

$$= \frac{8}{6} + 5 + 16 - 36 - 4 + 18 = -1 + \frac{8}{6} = \frac{2}{6} = \frac{1}{3}$$

2) Scalar double integral:

describe D as a simple region

$$D = \{(x,y) : 0 \leq y \leq 1, 0 \leq x \leq 2y\}$$

compute scalar curl of $\vec{F}$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - 1$

Integrate:

$$\int_0^1 \int_0^{2y} (2x-1) dx dy = \int_0^1 \left[x^2 - x\right]_{x=0}^{x=2y} dy$$

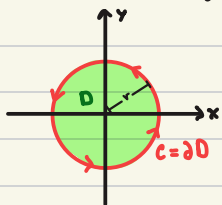
$$= \int_0^1 (4y^2 - 2y) dy = \left[\frac{4}{3}y^3 - y^2\right]_{y=0}^{y=1}$$

$$= \frac{4}{3} - 1 = \frac{1}{3}$$

Application of Green's Theorem

Sometimes computing a double integral might be easier,
other times computing a vector line integral might be easier

e.g. Compute $\oint_C -y dx + x dy$ where C is the circle of radius r centered at $(0,0)$ and oriented counter-clockwise



Instead of $\oint_C -y dx + x dy = \oint_C \vec{F} \cdot d\vec{s}$
let's compute the double integral:

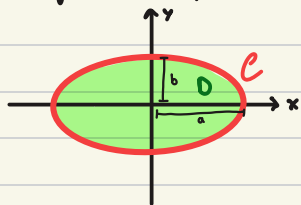
$$\vec{F}(x,y) = (-y, x)$$

$$\text{scalar curl: } \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 2$$

D is the disc of radius r

$$\oint_C -y dx + x dy = \iint_D 2 \, dA = 2 \, \text{Area}(D) = 2\pi r^2$$

e.g. Area enclosed by the ellipse C with eq. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



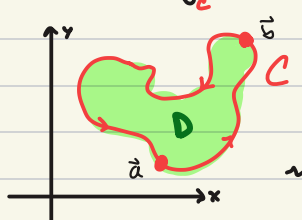
$$\text{We want } \text{Area}(D) = \iint_D 1 \, dx dy$$

Let me Green's Thm to compute this

Let $\vec{F}(x,y) = (\frac{-y}{2}, \frac{x}{2})$, scalar curl of $\vec{F}$ is 1.
Let's parametrize C : $\gamma(t) = (a \cos t, b \sin t)$ $\vec{\gamma}'(t) = \begin{pmatrix} -a \sin t \\ b \cos t \end{pmatrix}$
 $0 \leq t \leq 2\pi$

$$\begin{aligned} \text{Area}(D) &= \iint_D 1 \, dx dy = \oint_C \vec{F} \cdot d\vec{s} = \int_0^{2\pi} \vec{F}(\gamma(t)) \cdot \vec{\gamma}'(t) \, dt \\ &= \int_0^{2\pi} \left(-\frac{b \sin t}{2}, \frac{a \cos t}{2}\right) \cdot (-a \sin t, b \cos t) \, dt = \int_0^{2\pi} \frac{ab}{2} \sin^2 t + \frac{ab}{2} \cos^2 t \, dt \\ &= \frac{ab}{2} \int_0^{2\pi} \cos^2 t + \sin^2 t \, dt = \frac{ab}{2} (2\pi) = \pi(ab) \end{aligned}$$

e.g. Suppose $\vec{F} = \nabla f$ is a C^1 conservative v.f. in $\mathbb{R}^2$ and C is the oriented closed simple curve depicted. What is $\oint_C \vec{F} \cdot d\vec{s}$?



$f: \Sigma \subseteq \mathbb{R}^2 \xrightarrow{\text{open}} \mathbb{R}$ $C \subseteq \Sigma$
 $\vec{F} = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = \nabla f$
 scalar curl of $\vec{F}$: $\nabla \times \vec{F} = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = \frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x} = 0$

$$\oint_C \vec{F} \cdot d\vec{s} = \iint_D 0 \cdot dx dy = 0$$

This is true not just in $\mathbb{R}^2$, but in all $\mathbb{R}^n$.

vector line integral of conservative v.f.
 $\rightarrow$ are path independent

Theorem (Fundamental Theorem of Line Integrals, FTLI):

Let $f: \Sigma \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^1 function and $\vec{x}(t): [a, b] \rightarrow \Sigma \subseteq \mathbb{R}^n$ be a piecewise differentiable curve.

Then,

$$\int_{\vec{x}} \nabla f \cdot d\vec{s} = f(\vec{x}(t)) \Big|_{t=a}^{t=b} = f(\vec{x}(b)) - f(\vec{x}(a))$$

e.g. Work done by a conservative force (like gravity or electrostatic)

$\vec{F} = -\nabla E$
 force potential energy

Work: $W = \int_{\vec{x}} \vec{F} \cdot d\vec{s} = E_{\text{initial}} - E_{\text{final}} = \text{spent energy}$

The FTLI has very useful application, so leads to the question:
When is a vector field conservative?

Theorem: Suppose $\vec{F}: \Sigma \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a C^1 vector field, where
as usual we write $\vec{F}(x_1, \dots, x_n) = (F_1(x_1, \dots, x_n), F_2, \dots, F_n)$.

Then:

- 1) cff $\vec{F}$ is conservative, then $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ for all i, j in $1, \dots, n$.
 1*) cff for at least one pair i, j in $1, \dots, n$
 we have $\frac{\partial F_i}{\partial x_j} \neq \frac{\partial F_j}{\partial x_i}$, then $\vec{F}$ is not conservative.

- 2) cff all the mixed partial derivatives are equal
 (that is, $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ for all i, j in $1, \dots, n$) and
 Σ is convex (that is, for any points
 P and Q in Σ , the line segment from
 P to Q is contained inside Σ), then
 $\vec{F}$ is conservative.

e.g.-

